



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 14/08/2026

EXPERIMENT 21 -- MYSQL STRING FUNCTIONS- SPACE, SUBSTR, UPPER, LOWER, TRIM, LENGTH

AIM:

To implement MySQL String Functions such as **SPACE()**, **SUBSTR()**, **UPPER()**, **LOWER()**, **TRIM()**, **LENGTH()**, **BIT_LENGTH()**, **CHAR_LENGTH()**, **CONCAT()**, **INSTR()**, **LOCATE()**, **MID()**, and **POSITION()** using MySQL.

PROCEDURE:

1. Open MySQL Workbench and select your database.
2. Check the student_cgpa table.
3. Apply the required string functions to the student_name and department columns.
4. Execute each query and observe the output.
5. Verify the results of all string functions.

PROGRAM:

1. CREATE TABLE

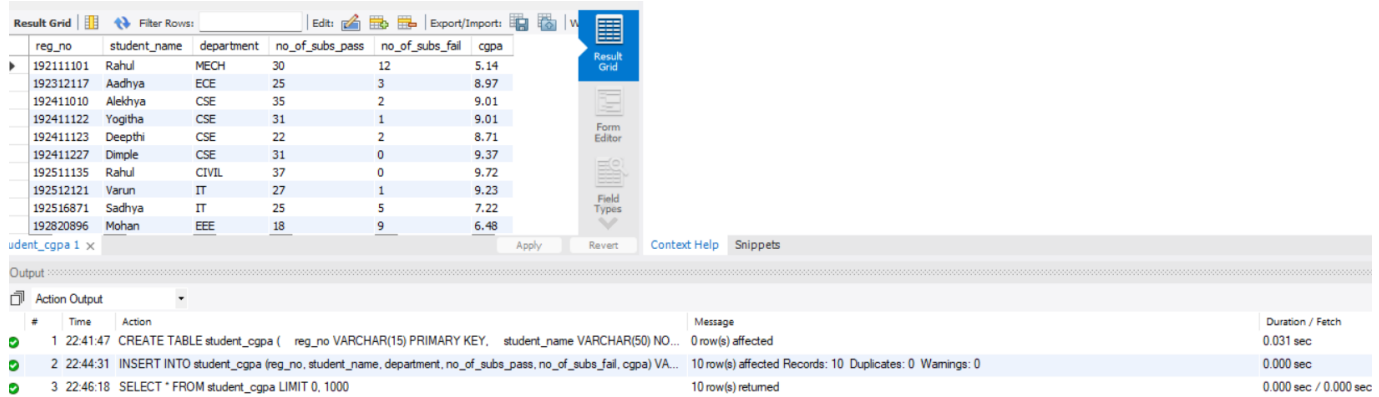
```
CREATE TABLE student_cgpa (  
    reg_no VARCHAR(15) PRIMARY KEY,  
    student_name VARCHAR(50) NOT NULL,  
    department VARCHAR(20),  
    no_of_subs_pass INT,  
    no_of_subs_fail INT,  
    cgpa DECIMAL(3,2)  
);
```

Then Insert Values:

```
INSERT INTO student_cgpa  
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)  
VALUES  
( '192411123', 'Deepthi', 'CSE', 22, 2, 8.71),  
( '192512121', 'Varun', 'IT', 27, 1, 9.23),  
( '192312117', 'Aadhya', 'ECE', 25, 3, 8.97),  
( '192411227', 'Dimple', 'CSE', 31, 0, 9.37),  
( '192820896', 'Mohan', 'EEE', 18, 9, 6.48),  
( '192411122', 'Yogitha', 'CSE', 31, 1, 9.01),  
( '192516871', 'Sadhya', 'IT', 25, 5, 7.22),  
( '192111101', 'Rahul', 'MECH', 30, 12, 5.14),  
( '192411010', 'Alekhya', 'CSE', 35, 2, 9.01),  
( '192511135', 'Rahul', 'CIVIL', 37, 0, 9.72);
```

```
SELECT * FROM student_cgpa;
```

OUTPUT:



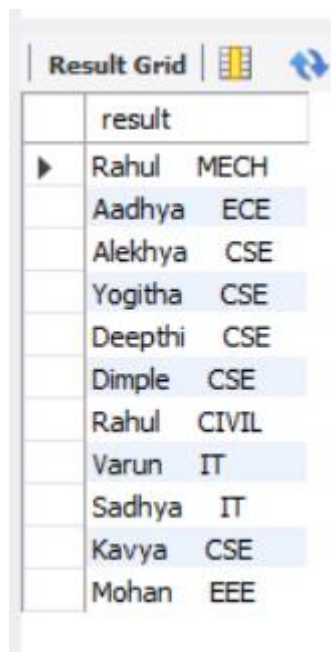
The screenshot shows a database management tool interface. At the top, there's a 'Result Grid' with columns: reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, and cgpa. Below the grid, there's an 'Output' section with a table showing execution logs.

#	Time	Action	Message	Duration / Fetch
1	22:41:47	CREATE TABLE student_cgpa (reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO...	0 row(s) affected	0.031 sec
2	22:44:31	INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	22:46:18	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

3. SPACE()

```
SELECT CONCAT(student_name, SPACE(5), department) AS result
FROM student_cgpa;
```

OUTPUT:



The screenshot shows a 'Result Grid' with a single column named 'result'. The data in this column is the concatenation of student names and departments, separated by five spaces.

result
Rahul MECH
Aadhya ECE
Alekhya CSE
Yogitha CSE
Deepthi CSE
Dimple CSE
Rahul CIVIL
Varun IT
Sadhya IT
Kavya CSE
Mohan EEE

4. SUBSTR()

```
SELECT student_name,
SUBSTR(student_name, 1, 3) AS substring_name
FROM student_cgpa;
```

OUTPUT:

Result Grid   Filter Rows:		
	student_name	substring_name
▶	Rahul	Rah
	Aadhya	Aad
	Alekhya	Ale
	Yogitha	Yog
	Deepthi	Dee
	Dimple	Dim
	Rahul	Rah
	Varun	Var
	Sadhya	Sad
	Kavya	Kav
	Mohan	Moh

5. UPPER()

```
SELECT student_name,
UPPER(student_name) AS upper_name
FROM student_cgpa;
```

OUTPUT:

Result Grid   Filter Rows:		
	student_name	upper_name
▶	Rahul	RAHUL
	Aadhya	AADHYA
	Alekhya	ALEKHYA
	Yogitha	YOGITHA
	Deepthi	DEEPTHI
	Dimple	DIMPLE
	Rahul	RAHUL
	Varun	VARUN
	Sadhya	SADHYA
	Kavya	KAVYA
	Mohan	MOHAN

6. LOWER()

```
SELECT student_name,
LOWER(student_name) AS lower_name
FROM student_cgpa;
```

OUTPUT:

Result Grid   Filter Rows:		
	student_name	lower_name
▶	Rahul	rahul
	Aadhya	aadhya
	Alekhya	alekhya
	Yogitha	yogitha
	Deepthi	deepthi
	Dimple	dimple
	Rahul	rahul
	Varun	varun
	Sadhya	sadhya
	Kavya	kavya
	Mohan	mohan

7. TRIM()

```
SELECT student_name,
       TRIM(CONCAT(' ', student_name, ' ')) AS trimmed_name
FROM student_cgpa;
```

OUTPUT:

Result Grid   Filter Rows:		
	student_name	trimmed_name
▶	Rahul	Rahul
	Aadhya	Aadhya
	Alekhya	Alekhya
	Yogitha	Yogitha
	Deepthi	Deepthi
	Dimple	Dimple
	Rahul	Rahul
	Varun	Varun
	Sadhya	Sadhya
	Kavya	Kavya
	Mohan	Mohan

8. LENGTH()

```
SELECT student_name,
       LENGTH(student_name) AS name_length
FROM student_cgpa;
```

OUTPUT:

Result Grid   Filter Rows:		
	student_name	name_length
▶	Rahul	5
	Aadhya	6
	Alekhya	7
	Yogitha	7
	Deepthi	7
	Dimple	6
	Rahul	5
	Varun	5
	Sadhya	6
	Kavya	5
	Mohan	5

9. BIT_LENGTH()

```
SELECT student_name,
       BIT_LENGTH(student_name) AS bit_length
FROM student_cgpa;
```

OUTPUT:

Result Grid   Filter Rows:		
	student_name	bit_length
▶	Rahul	40
	Aadhya	48
	Alekhya	56
	Yogitha	56
	Deepthi	56
	Dimple	48
	Rahul	40
	Varun	40
	Sadhya	48
	Kavya	40
	Mohan	40

10. CHAR_LENGTH()

```
SELECT student_name,
       CHAR_LENGTH(student_name) AS character_length
FROM student_cgpa;
```

OUTPUT:

Result Grid			Filter Rows:
	student_name	character_length	
▶	Rahul	5	
	Aadhya	6	
	Alekhya	7	
	Yogitha	7	
	Deepthi	7	
	Dimple	6	
	Rahul	5	
	Varun	5	
	Sadhya	6	
	Kavya	5	
	Mohan	5	

11. CONCAT()

```
SELECT CONCAT(student_name, ' - ', department) AS student_details
FROM student_cgpa;
```

OUTPUT:

Result Grid		Filter
	student_details	Resets all
▶	Rahul - MECH	
	Aadhya - ECE	
	Alekhya - CSE	
	Yogitha - CSE	
	Deepthi - CSE	
	Dimple - CSE	
	Rahul - CIVIL	
	Varun - IT	
	Sadhya - IT	
	Kavya - CSE	
	Mohan - EEE	

12. INSTR()

```
SELECT student_name,
INSTR(student_name, 'a') AS position_of_a
FROM student_cgpa;
```

OUTPUT:

Result Grid			Filter Rows:
	student_name	position_of_a	
▶	Rahul	2	
	Aadhya	1	
	Alekhya	1	
	Yogitha	7	
	Deepthi	0	
	Dimple	0	
	Rahul	2	
	Varun	2	
	Sadhya	2	
	Kavya	2	
	Mohan	4	

13. LOCATE()

```
SELECT student_name,
LOCATE('a', student_name) AS position_of_a
FROM student_cgpa;
```

OUTPUT:

Result Grid			Filter Rows:
	student_name	position_of_a	
▶	Rahul	2	
	Aadhya	1	
	Alekhya	1	
	Yogitha	7	
	Deepthi	0	
	Dimple	0	
	Rahul	2	
	Varun	2	
	Sadhya	2	
	Kavya	2	
	Mohan	4	

14. MID()

```
SELECT student_name,
MID(student_name, 2, 3) AS middle_characters
FROM student_cgpa;
```

OUTPUT:

Result Grid			Filter Rows:
	student_name	position_of_a	
▶	Rahul	2	
	Aadhya	1	
	Alekhya	1	
	Yogitha	7	
	Deepthi	0	
	Dimple	0	
	Rahul	2	
	Varun	2	
	Sadhya	2	
	Kavya	2	
	Mohan	4	

15. POSITION()

```
SELECT student_name,
       POSITION('a' IN student_name) AS position_of_a
FROM student_cgpa;
```

OUTPUT:

Result Grid			Filter Rows:
	student_name	position_of_a	
▶	Rahul	2	
	Aadhya	1	
	Alekhya	1	
	Yogitha	7	
	Deepthi	0	
	Dimple	0	
	Rahul	2	
	Varun	2	
	Sadhya	2	
	Kavya	2	
	Mohan	4	

RESULT: Thus, the MySQL String Functions such as **REPLACE**, **REPEAT**, **REVERSE**, **RIGHT**, **LEFT**, **RPAD**, **LPAD**, **ASCII**, **BIN**, and **OCT** were successfully implemented using the student_cgpa table.